

ECON 1550

Spring 2026

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Submission: Canvas or Gradescope

Problem Set 7

Due: April 1, 2026 at 11:59pm ET

Instructions

- When submitting to Gradescope, indicate the page where each question is answered to avoid a 5-point deduction.
- Full credit is given for correct answers. If multiple steps are needed, you must show them to get full credit.
- Points are shown for each part. Partial credit is given for partially correct answers; show your work to maximize it.
- Late submissions receive a score of zero.
- If you have technical problems submitting, email your work to the Head TA before the deadline.
- Collaboration with classmates is encouraged; use of generative AI is permitted but discouraged.
- You must write, understand, and submit your solutions individually. Copying other students' or AI-generated answers, even fragments, is not allowed.

1. Chapter 6: Output and the Exchange Rate in the Short Run (20 points)

The following questions analyze fiscal and monetary policy using the *AA-DD* model.

- (a) [5 points] If a government initially has a balanced budget but then cuts taxes, it is running a deficit. Suppose the government finances its deficit by printing the extra money it now needs to cover its expenditures, that is, money supply increases by the same amount as the deficit. According to the *AA-DD* model, how does the nominal exchange rate respond to this simultaneous change in taxes and money supply in the short run?
- (b) [5 points] A new government is elected and announces that once it is inaugurated, it will permanently increase the money supply. Use the *AA-DD* model to study the economy's response to this announcement.

During the last few years, and especially after Covid, there have been many calls to “buy American”. Imagine the government implements a “buy American” program in which

any new government spending is constrained to only purchase domestic goods. Use the AA-DD model to answer the following questions:

- (c) [5 points] Does a *permanent* increase in U.S. government spending constrained by “buy American” restrictions have a bigger effect on U.S. output than unconstrained U.S. government spending? Why or why not? Make sure you consider both the short and long run effects on output.
- (d) [5 points] Now assume the government spending is temporary. Do “buy American” restrictions have a bigger effect on U.S. output than unconstrained U.S. government spending? Why or why not? Make sure you consider both the short and long run effects on output.

2. Import Tariffs in the AA-DD Model (80 points)

Consider a small open economy. Use the notation E for the nominal exchange rate, q for the real exchange rate, Y for output, Y^d for disposable income, R for the domestic nominal interest rate, and R^* for the foreign nominal interest rate. The real exchange rate is

$$q \equiv \frac{EP^*}{P}.$$

A rise in E is a nominal depreciation, and a rise in q is a real depreciation. The standard model without tariffs is

$$\begin{aligned} Y^d &= Y - T, \\ C &= C(Y^d), \\ EX &= EX(q, Y^*), \\ IM &= qIM^*(q, Y^d), \\ CA &= EX - IM, \\ D &= C + I + G + CA, \\ Y &= D, \\ \frac{M^s}{P} &= L(R, Y), \\ R &= R^* + \frac{E^e}{E} - 1. \end{aligned}$$

Assume C is increasing in Y^d . Assume EX is increasing in q and in Y^* . Assume IM^* is decreasing in q and increasing in Y^d . Assume L is decreasing in R and increasing in Y . For the short-run AA-DD block, take $T, I, G, M^s, P, P^*, R^*, E^e$, and Y^* as exogenous. The

short-run endogenous variables are Y , Y^d , C , EX , IM^* , IM , CA , D , R , E , and q . Use the following functional forms:

$$\begin{aligned} C(Y^d) &= \frac{1}{2}Y^d, \\ EX(q, Y^*) &= \frac{1}{4} + \frac{1}{10}q + \frac{1}{10}(Y^* - 1), \\ IM^*(q, Y^d) &= \frac{5 + 2Y^d}{20q} - \frac{1}{10}, \\ L(R, Y) &= \frac{Y}{1 + R}. \end{aligned}$$

For parts (a) through (c), use

$$T = 0, \quad I = \frac{1}{5}, \quad G = \frac{1}{5}, \quad M^s = 1, \quad P^* = 1, \quad R^* = 0, \quad Y^* = 1.$$

Keep the current domestic price level P and the expected future exchange rate E^e symbolic until you are asked to solve an equilibrium. The numerical choices above are made so that all equilibrium values in the problem stay economically meaningful and positive.

- (a) [6 points] Recall that value imports equal the relative price of imports times import volume. Using the functional form above, explain the price effect and the volume effect of an increase in q . Then verify that in this numerical model the volume effect dominates, so value imports are decreasing in q . Do not use the phrase ‘Marshall-Lerner condition’.
- (b) [6 points] Derive the DD curve explicitly. Explain what points on the DD curve represent. In a graph with E on the vertical axis and Y on the horizontal axis, is the DD curve increasing or decreasing. Give intuition for that sign.
- (c) [6 points] Repeat part (b) for the AA curve.
- (d) [6 points] For the remaining parts of the question, add a Phillips curve. Set expected inflation equal to zero and write the Phillips curve using the output gap $Y - Y^f$. Denote the slope by α . State clearly which variables are endogenous and which are exogenous in this Phillips-curve block. Then set $\alpha = 1$ and $Y^f = 1$. Then solve for the initial long-run equilibrium of the no-tariff model.
- (e) [7 points] Now add an exogenous ad valorem import tariff τ . The tariff raises the domestic-currency price of imported goods from q to $(1 + \tau)q$. Replace the import-volume function by

$$IM^* = IM^*((1 + \tau)q, Y^d),$$

so border-value imports are

$$IM = qIM^*((1 + \tau)q, Y^d).$$

For the remaining parts, use $\bar{\tau} = 1/3$. For the temporary case, the tariff is higher only in the short run, so $\tau_{SR} = \bar{\tau}$ and $\tau_{LR} = 0$. For the permanent case, $\tau_{SR} = \tau_{LR} = \bar{\tau}$. At the date of the shock, the short-run price level is fixed at its initial value, so $P_{SR} = P_0$. Keeping the real exchange rate and output fixed, how does the tariff affect the current account.

- (f) [7 points] Find the DD curve when the tariff is present. How do tariffs change the DD curve. Comment on both the slope and the intercept. Give intuition.
- (g) [7 points] Find the AA curve. How do tariffs change the AA curve. Comment on the slope and the intercept if that language is useful. Give intuition.
- (h) [7 points] Assume the economy starts with $\tau_0 = 0$. Solve for the initial long-run equilibrium in the tariff model.
- (i) [7 points] Consider a temporary increase in the tariff from 0 to $\bar{\tau} = 1/3$. Sketch in an AA-DD diagram the initial equilibrium, the short-run equilibrium, the long-run equilibrium, and the transition between the short run and the long run. Explain in words how and why the curves shift, or do not shift, over time. A schematic sketch is enough.
- (j) [7 points] Plot the time paths of E , q , Y , P , inflation, CA , EX , and IM for the temporary tariff increase. Use 0 for the initial long-run equilibrium and SR for the short-run equilibrium, and mark the later date at which the temporary tariff is removed. A schematic path is enough after SR .
- (k) [7 points] Repeat part (i) for a permanent increase in the tariff from 0 to $\bar{\tau} = 1/3$.
- (l) [7 points] Repeat part (j) for the permanent tariff increase. Use 0 for the initial long-run equilibrium and SR for the short-run equilibrium.